

One image $\rightarrow$ millions of images

- $\Rightarrow$ NN b'ni rasmlar orqali train qilamiz
- $\Rightarrow$ Image'lar k'p b'liqlik kerak
- $\Rightarrow$ Har bir image t'pni yuklanishi kerak
- $\Rightarrow$ Har bir image label qo'yilishi kerak
target qiymat (classification)
- $\Rightarrow$ Har bir image bir xil formatda b'liqlik k-k.

Tabiiki train uchun o'zgar b'ladir chunki 1 ta image bu emmas millionlab image'lar bir train qilinishi kerak.

1 image $\Rightarrow$ juda oz o'rganish
many images $\Rightarrow$ better learning

Image'ni dog yoki cat t'pini aniqlayotirgan (classification task) yaratyabmiz

$\Rightarrow$ Image b'ni (juda ham k'p image'lar b'di) folder'larga joylab olamiz (tartibli)

$\Rightarrow$ train va test folder o'chiramiz
ichiga cat va dog uchun alohida folder'lar ochib image'larini joylaymiz

<u>Labelled</u>	$\rightarrow$ cat $\rightarrow$ dog
<u>Output</u>	$\begin{matrix} \checkmark & \times \\ \text{cat} & \text{cat} \end{matrix} \rightarrow \text{dog}$

"dog" folderda "cat"
oe aksincha bo'lmaydi
kerak

(Dataset): collection of samples used for training

(Data + labels)

labels $\rightarrow$ output

Image Classification:

input $\rightarrow$ image $\rightarrow$ cat photo

output $\rightarrow$ label $\rightarrow$ cat

Hər bir image ni labeləli çıxış

kerək. Bu cat image yoxi dog

image olub. Oldu qatə qitinyan

hozir avtomat kallyan

(Autolabeling)

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(Manual labeling)

SML $\Rightarrow$ data preprocessing. (1)

Oh $\Rightarrow$ image processing $\rightarrow$ (Transforms)

image data bilan illayt paxanə
(data preprocessing)

Images real world:

turli xil olcham.

turli xil format (jpg, png)

turli xil pixel

Model

bir xil olcham

bir xil format

bir xil pixel

}

kutadi

Transform:

Resize - Rasmlarni turli shtamda bo'lsa

Crop - Faqat muhim qismini ajratib olib k..k bo'lsa.

Rotate/Flip - Modelni ko'proq variantlar bilan o'qitish uchun.

Grayscale - Rang muhim bo'lmaganida

Array conversion - Rasmlni satalarga aylantirish uchun

Normalization - Modelni barqaror va tez o'qitish uchun.